



Hybrid Multimodal Modelling for Joint Chip Morphology Classification and Tool Wear Prediction in Computer Numerical Control Milling

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Abstract: Reliable monitoring of chip formation and tool degradation is essential for machining quality, process stability, and condition-based maintenance in intelligent manufacturing systems. However, individual sensing modalities and isolated learning models provide incomplete representations of the heterogeneous and time-dependent interactions that characterize computer numerical control (CNC) milling. This study investigates a hybrid multimodal modelling framework that integrates heterogeneous sensing, complementary feature-learning mechanisms, and joint classification–regression tasks for chip morphology and progressive flank-wear monitoring. Vibration, acoustic emission (AE), cutting force, spindle current, and temperature signals were synchronized and fused within a unified modelling architecture. A convolutional neural network (CNN) was used to extract local signal characteristics, a bidirectional long short-term memory (BiLSTM) network modelled temporal dependencies, and Transformer-based self-attention captured long-range interactions within the fused representation. The resulting features were used jointly to classify continuous, segmented, discontinuous, and serrated chip morphologies and to estimate progressive flank wear. The framework was evaluated under variations in spindle speed, feed rate, and wet and dry machining conditions, together with sensor-fusion comparisons and architectural ablation experiments. The full multimodal configuration achieved approximately 97.2% chip-classification accuracy, while the complete hybrid architecture reached approximately 98% in the ablation analysis and maintained prediction accuracy above 94% across the investigated cutting conditions. Tool-wear estimates also showed close agreement with the experimentally measured values. These results indicate that coordinated integration of heterogeneous sensing with complementary local, sequential, and attention-based modelling mechanisms improves predictive stability and robustness under variable machining conditions. The proposed framework provides a structured hybrid-modelling approach for intelligent machining-process monitoring and offers a basis for predictive maintenance and adaptive CNC manufacturing systems.

Keywords: Hybrid modelling; Multimodal sensor fusion; Computer numerical control milling; Tool wear prediction; Chip morphology classification; Intelligent machining systems

1 Introduction

The continuing development of intelligent manufacturing and Industry 4.0 has increased the demand for automated machining-process monitoring, intelligent tool-condition assessment, and reliable chip-morphology analysis. In computer numerical control (CNC) milling, early monitoring approaches were largely based on empirical machining theories, analytical force models, and statistical signal-processing techniques, with vibration and cutting-force measurements widely used to detect tool degradation and machining instability [1]. Although these approaches can provide reliable information under controlled laboratory conditions, their performance may deteriorate as cutting parameters, workpiece materials, or environmental conditions change. Conventional feature-engineering methods also depend heavily on domain expertise and may not adequately represent the nonlinear interactions between machining parameters and heterogeneous sensor signals [2].

Machine-learning techniques, including support vector machines (SVMs), random forests (RFs), and extreme gradient boosting (XGBoost), have increasingly been applied to intelligent tool-condition monitoring to overcome some of these limitations. Such methods can improve predictive performance by learning latent relationships

between machining states and sensor-derived features from experimental data [3, 4]. Nevertheless, conventional machine-learning approaches generally remain dependent on manually designed features, which restricts their ability to represent high-dimensional, heterogeneous, and temporally varying machining signals. Deep-learning methods offer an alternative by allowing more informative representations to be learned directly from raw or minimally processed sensor data.

Convolutional neural networks (CNNs) have consequently been adopted for machining-signal analysis because of their ability to extract local patterns and correlations automatically from multidimensional signal representations [5]. They have been applied to tool-wear monitoring, chatter detection, and machining-fault diagnosis. Their strength in local feature extraction, however, does not directly address the long-range temporal dependencies inherent in sequential machining data. Recurrent architectures, particularly long short-term memory (LSTM) and bidirectional LSTM (BiLSTM) networks, provide a complementary modelling mechanism for temporal relationships in vibration, acoustic emission (AE), and other sensor signals [6]. Combining CNN-based feature extraction with LSTM-based temporal modelling has shown improved performance in progressive tool-wear monitoring [7]. Recurrent processing nevertheless becomes increasingly demanding for long sequences and offers limited parallelism, leaving scope for modelling structures capable of capturing broader temporal interactions.

Transformer architectures provide such a mechanism through self-attention, which models relationships between distant elements of a sequence without relying exclusively on recurrent processing. Their capacity to assign adaptive importance to informative features and capture long-range dependencies has led to applications in industrial time-series analysis, predictive maintenance, anomaly detection, and intelligent fault diagnosis [8, 9]. Their role in the simultaneous analysis of chip morphology and tool wear in CNC milling, however, remains comparatively underexplored. More importantly, the problem extends beyond the use of a particular learning architecture: it concerns how complementary spatial, sequential, and long-range representations can be coordinated with heterogeneous machining signals within a unified modelling structure. Whether such integration can improve the joint representation of chip formation and progressive tool degradation under changing machining conditions remains an open question.

Multimodal sensor fusion offers a natural basis for this integration. CNC milling processes generate heterogeneous information through vibration, AE, cutting force, spindle current, and temperature measurements. These sensing modalities describe different aspects of machining dynamics and, when considered jointly, can provide a more comprehensive process representation while reducing sensitivity to noise or disturbances affecting individual sensors [10]. Multimodal learning has accordingly been reported to produce more reliable predictions than approaches based on individual sensing modalities [11]. Yet the benefit of additional sensing channels depends strongly on how their information is coordinated within the modelling architecture. Simple concatenation or relatively shallow fusion may not sufficiently represent interactions across sensor channels, temporal scales, and machining states. A structured hybrid framework is therefore required in which heterogeneous sensing is coupled with complementary modelling components rather than treated merely as an expanded input space.

Chip morphology provides an important target for such integrated monitoring because chip formation reflects cutting stability, heat generation, material deformation, and tool-workpiece interactions. Depending on the cutting conditions and material response, chips can exhibit continuous, segmented, serrated, or discontinuous morphologies, with each form associated with distinct characteristics in the acquired sensor signals [12]. Existing machine-learning approaches to chip classification have primarily relied on image-based analysis or individual sensor modalities. Chip morphology has also commonly been treated as an independent classification problem, despite its physical relationship with the progressive degradation of the cutting tool. Separating these two monitoring tasks can therefore overlook information shared between chip-formation behaviour and tool condition. Their joint treatment within a common multimodal representation provides an opportunity to model complementary manifestations of the same evolving machining process.

A further challenge is maintaining reliable performance as machining conditions vary. Industrial milling is conducted across different spindle speeds, feed rates, lubrication conditions, and operating environments. Models trained within a restricted operating regime can experience substantial performance degradation when exposed to previously unseen conditions [13]. Robustness to such variations is therefore essential for practical deployment. It is also an important test of modelling integration: a hybrid architecture should not merely improve performance under a fixed experimental setting but should preserve useful representations when the operating state or the reliability of individual sensing channels changes. Systematic evaluation under variable and out-of-distribution machining conditions nevertheless remains limited, while interpretability, computational efficiency, and effective coordination of heterogeneous sensor information continue to present challenges for deep-learning-based machining monitoring.

Several related gaps follow from the preceding discussion. First, tool-wear monitoring and chip-morphology classification have largely been investigated as separate tasks, despite their coupled evolution during milling. Second, conventional machine-learning approaches continue to rely substantially on handcrafted features, limiting their capacity to construct adaptive representations from complex machining signals [14]. Third, many deep-

learning frameworks do not fully exploit interactions among heterogeneous sensing channels or explicitly coordinate modelling mechanisms operating at different representational and temporal scales. Fourth, although Transformer-based architectures have demonstrated considerable potential for industrial time-series modelling, their structured integration with multimodal sensing and complementary local and sequential models for joint chip/tool analysis remains limited. Finally, the robustness of such integrated models under changes in spindle speed, feed rate, and lubrication condition has not been sufficiently established [15]. Recent studies have begun to address these limitations through more integrated sensing and modelling strategies. Multisensor fusion architectures have been developed to combine force, vibration, and AE information for tool-wear estimation, demonstrating the value of integrating heterogeneous process observations at multiple representation levels [16, 17]. More recent approaches have incorporated multimodal information fusion with deep representation learning and domain adaptation to improve tool-wear prediction across varying machining conditions [18]. Hierarchical temporal-modal fusion has also been investigated to explicitly model time-varying correlations among multimodal monitoring signals [19], while multimodal neural architectures have been extended towards interpretable tool-wear assessment [20]. Beyond tool-wear estimation alone, sensor-based machine-learning frameworks have demonstrated the feasibility of predicting chip morphology from heterogeneous machining signals [21]. Collectively, these developments indicate a shift from isolated predictors towards integrated multimodal monitoring architectures. However, the coordinated use of local feature extraction, bidirectional temporal modelling, and long-range self-attention within a unified framework for the joint analysis of chip morphology and progressive tool wear remains insufficiently investigated. These limitations point to the need for a hybrid modelling architecture in which heterogeneous sensor fusion, local feature extraction, temporal sequence modelling, and long-range attention are coordinated within a unified engineering-monitoring framework.

To address these gaps, this study develops a hybrid multimodal modelling framework for the joint classification of chip morphology and prediction of progressive tool wear in CNC milling. The framework integrates synchronized vibration, AE, cutting-force, spindle-current, and temperature signals to represent complementary aspects of the machining state. Rather than relying on a single modelling paradigm, the architecture coordinates three complementary learning mechanisms: a CNN extracts localized characteristics from the sensor representation, a BiLSTM models bidirectional temporal dependencies, and Transformer-based self-attention captures longer-range interactions within the learned sequence. The integrated representation supports two related monitoring tasks: classification of continuous, segmented, discontinuous, and serrated chip morphologies and regression-based estimation of progressive flank wear. The resulting architecture therefore treats sensor fusion, representation learning, temporal modelling, and joint condition assessment as interconnected components of a single intelligent machining system.

The main contributions of this study are as follows:

- A hybrid multimodal modelling framework is developed for the joint analysis of chip morphology and progressive tool wear, integrating heterogeneous machining signals within a unified classification-regression architecture;
- A coordinated CNN-BiLSTM-Transformer structure combines local feature extraction, bidirectional temporal modelling, and long-range self-attention, enabling complementary characteristics of the machining process to be represented at different temporal and feature scales;
- Multimodal sensor fusion is evaluated across variations in spindle speed, feed rate, and wet and dry milling conditions to determine whether the integrated modelling structure maintains stable predictive behaviour under different machining environments;
- The framework is experimentally evaluated under the specified machining conditions to examine its applicability to intelligent machining monitoring beyond a single operating configuration;
- Comparative and ablation analyses against conventional machine-learning and deep-learning approaches are conducted to distinguish the contributions of sensing modalities and architectural components to chip-morphology classification and tool-wear prediction.

Together, these contributions shift the emphasis from isolated algorithmic improvement towards the structured integration of heterogeneous sensing and complementary modelling mechanisms. The study thereby examines not only predictive performance, but also how hybrid modelling architecture influences robustness and operational monitoring capability in an intelligent CNC milling system.

2 Proposed Methodology

A hybrid multimodal modelling framework was developed for the joint classification of chip morphology and prediction of progressive tool wear in CNC milling. The framework was designed to integrate heterogeneous sensing information with complementary feature-learning and temporal-modelling mechanisms within a unified engineering-monitoring architecture. Vibration, AE, cutting force, spindle current, and temperature signals were first acquired and preprocessed to obtain synchronized representations of the machining process. Local signal characteristics

were then extracted through convolutional modelling, bidirectional temporal dependencies were represented by a BiLSTM network, and longer-range interactions were captured through Transformer-based self-attention. The resulting multimodal representation was subsequently used for both chip-morphology classification and flank-wear prediction.

The methodological design therefore combines five interconnected modelling functions: heterogeneous sensor acquisition, signal preprocessing and feature representation, multimodal feature integration, hybrid temporal modelling, and joint classification–regression output. Rather than treating these components as independent processing stages, the framework coordinates them so that information describing cutting dynamics, thermal behaviour, machine response, chip formation, and tool degradation can be represented within a common modelling structure.

2.1 Overall Framework Architecture

The proposed framework consists of five major stages:

- (1) Multimodal sensor acquisition;
- (2) Signal preprocessing and feature extraction;
- (3) Deep multimodal feature learning;
- (4) Transformer-based temporal attention modeling;
- (5) Chip classification and tool wear prediction.

The complete modelling workflow is illustrated in Figure 1. Heterogeneous machining signals are first acquired from multiple sensing channels and subjected to filtering, normalization, window segmentation, and statistical processing. Features derived from the different modalities are then integrated into a common multimodal representation. This representation is processed through a hybrid CNN–BiLSTM–Transformer architecture, in which the CNN captures localized patterns, the BiLSTM represents bidirectional temporal dependencies, and the Transformer models longer-range interactions through self-attention. The integrated representation is finally supplied to separate classification and regression outputs for chip-morphology recognition and progressive flank-wear estimation, respectively.

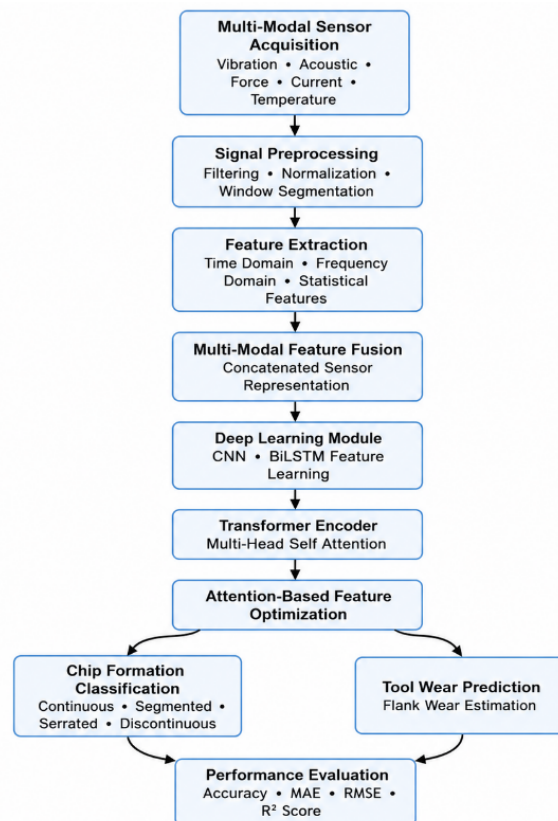


Figure 1. Overall architecture of the proposed hybrid multimodal modelling framework for computer numerical control (CNC) milling monitoring

Note: CNN: convolutional neural network; BiLSTM: bidirectional long short-term memory; MAE: mean absolute error; RMSE: root mean square error.

2.2 Multimodal Sensor Data Acquisition

Multiple sensing modalities were used to characterize CNC milling behaviour under different cutting conditions. Because no individual sensor provides a complete representation of the machining state, signals associated with mechanical vibration, acoustic activity, cutting forces, electrical loading, and thermal behaviour were acquired simultaneously. The sensing channels comprised:

- Vibration signals;
- AE signals;
- Cutting-force signals;
- Spindle motor current;
- Temperature measurements.

These signals describe complementary aspects of the milling process. Vibration and AE measurements contain information related to dynamic instability and transient cutting events, cutting-force signals reflect tool–workpiece interaction, spindle current provides an indirect measure of machining load, and temperature measurements characterize the thermal state associated with friction, chip formation, and progressive tool degradation. Their coordinated use therefore provides a heterogeneous representation of machining behaviour that cannot be obtained from a single sensing modality.

Let the acquired multimodal sensor data be represented as follows:

$$X = x_v, x_a, x_f, x_c, x_t \quad (1)$$

where:

- $x_v \rightarrow$ vibration signal;
- $x_a \rightarrow$ acoustic signal;
- $x_f \rightarrow$ force signal;
- $x_c \rightarrow$ current signal;
- $x_t \rightarrow$ temperature signal.

The resulting dataset contained synchronized sensor observations together with chip-morphology labels and experimentally measured tool-wear values obtained under different spindle speeds, feed rates, and lubrication conditions. These data formed the basis for both classification and regression modelling.

2.3 Signal Preprocessing

Raw machining signals can contain measurement noise, transient disturbances, baseline variation, and sensor-specific inconsistencies. Preprocessing was therefore performed before feature extraction and hybrid modelling to reduce unwanted variation and provide comparable input representations across sensing modalities.

The preprocessing pipeline consisted of:

- Low-pass filtering;
- Signal normalization;
- Window segmentation;
- Statistical smoothing.

Low-pass filtering and smoothing were applied to suppress high-frequency noise and short-term fluctuations that were not associated with the underlying machining state. Because the sensing channels differed in scale and physical units, each signal was subsequently normalized using min–max normalization:

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (2)$$

where:

- x is the original sensor value;
- $x_{\min}$ and $x_{\max}$ are the minimum and maximum values, respectively, of the signal.

The normalized signals were then divided into fixed-length windows. Window-based segmentation enabled local and temporal characteristics to be represented consistently across the multimodal inputs and provided sequential samples suitable for CNN, BiLSTM, and Transformer-based modelling.

2.4 Feature Extraction

Features were extracted from each sensor modality to characterize the machining process in both the time and frequency domains. This step was intended to preserve physically meaningful information associated with signal magnitude, fluctuation, distributional shape, impulsiveness, and spectral behaviour before multimodal integration.

The extracted statistical features included:

- Mean;

- Root mean square (RMS);
- Kurtosis;
- Skewness;
- Variance;
- Peak amplitude;
- Spectral energy.

Among these descriptors, RMS provides a measure of overall signal energy and was calculated as follows:

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2} \quad (3)$$

where:

- N denotes the number of signal samples;
- x_i represents the sensor amplitude of sample i .

The feature representations derived from the individual sensor channels were subsequently combined to construct a multimodal description of the machining state. This representation retained information from different physical processes while providing a common input space for the subsequent hybrid modelling stages.

2.5 Multimodal Feature Fusion

Feature-level fusion was employed to integrate complementary information from the heterogeneous sensing modalities. Rather than analysing vibration, AE, cutting force, spindle current, and temperature independently, the modality-specific feature representations were concatenated to form a unified multimodal feature vector.

The fused feature representation is expressed as follows:

$$F_{\text{fusion}} = [F_v \oplus F_a \oplus F_f \oplus F_c \oplus F_t] \quad (4)$$

where:

- $F_v \rightarrow$ vibration features;
- $F_a \rightarrow$ acoustic features;
- $F_f \rightarrow$ force features;
- $F_c \rightarrow$ current features;
- $F_t \rightarrow$ temperature features;
- $\oplus$ denotes feature concatenation.

This fusion structure enables complementary information from different physical sensing processes to be represented jointly. Signals that are sensitive to different aspects of machining behaviour can therefore compensate for limitations in individual modalities, while the subsequent learning architecture can model interactions among the fused features. This integration is particularly important under varying cutting conditions, where the reliability or informativeness of a single sensor may change.

2.6 Hybrid Deep Learning Architecture

The fused multimodal representation was processed using a hybrid CNN–BiLSTM architecture before Transformer-based attention modelling. The two components perform complementary functions: the CNN extracts localized characteristics from the multimodal sequence, whereas the BiLSTM represents temporal dependencies in both forward and backward directions.

2.6.1 Convolutional neural network-based local feature extraction

A one-dimensional CNN was used to extract localized patterns from the preprocessed multimodal sequences. Convolutional operations allow short-range variations and characteristic signal structures associated with chip formation, cutting instability, and tool degradation to be represented automatically.

The convolution operation is defined as follows:

$$y_i = \sum_{k=1}^K w_k x_{i-k} + b \quad (5)$$

where:

- w_k denotes the convolution kernel weights;
- b is the bias term;
- K represents the kernel size.

Through shared convolutional filters, the CNN transforms the fused sensor representation into higher-level local features. These features provide a compact representation of short-range machining behaviour and serve as the input to subsequent temporal modelling.

2.6.2 Bidirectional long short-term memory-based temporal modelling

Machining signals evolve sequentially as chip formation, thermal behaviour, cutting loads, and tool wear change over time. A BiLSTM layer was therefore employed to model temporal dependencies in both directions within the extracted feature sequences.

The hidden-state update is expressed as follows:

$$h_t = f(W_h h_{t-1} + W_x x_t + b) \quad (6)$$

where:

- x_t denotes the input sequence;
- h_t represents the hidden state;
- W_h and W_x are trainable weights.

The bidirectional structure allows information preceding and following a given sequence position to contribute to its temporal representation. This is useful for progressive tool-wear modelling and chip-state recognition because current sensor patterns may depend on both preceding process evolution and neighbouring temporal observations.

2.7 Transformer-Based Self-Attention

Although the CNN and BiLSTM capture local and sequential dependencies, machining signals may also contain relationships that extend across longer temporal intervals. A multi-head self-attention Transformer was therefore incorporated after the preceding feature-learning stages to capture long-range interactions within the multimodal representation.

The self-attention operation is defined as follows:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \quad (7)$$

where:

- $Q \rightarrow$ query matrix;
- $K \rightarrow$ key matrix;
- $V \rightarrow$ value matrix;
- $d_k \rightarrow$ key dimensionality.

The self-attention mechanism determines the relative importance of different positions within the learned sequence by evaluating interactions among the query, key, and value representations. Multiple attention heads allow different temporal and feature relationships to be represented simultaneously. The Transformer therefore complements CNN-based local modelling and BiLSTM-based sequential modelling by capturing dependencies that span longer portions of the machining process.

Within the overall hybrid architecture, the CNN, BiLSTM, and Transformer are not treated as interchangeable predictors. Each component addresses a different representational requirement, and their sequential coordination forms the central integration logic of the proposed modelling framework.

2.8 Chip-Morphology Classification

The final Transformer-derived representation was supplied to a softmax classification head to determine the chip morphology associated with each machining condition.

The softmax function is given as follows:

$$P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}} \quad (8)$$

where:

- z_i represents class logits;
- C denotes the number of chip categories.

Four chip-morphology classes were considered:

- Continuous chips;
- Segmented chips;
- Discontinuous chips;
- Serrated chips.

The classification task therefore maps the integrated multimodal representation onto discrete chip-formation states. Because the classifier operates on features shared with the tool-wear prediction branch, chip morphology is analysed within the same machining-state representation rather than as an independent sensing problem.

2.9 Tool-Wear Prediction

A regression head was constructed using the fused Transformer representation to estimate progressive flank wear. In contrast to chip morphology, which is represented as a discrete classification problem, tool wear was treated as a continuous condition indicator.

The regression loss function used for model optimization is expressed as follows:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (9)$$

where:

- y_i is the experimentally measured tool-wear value;
- $\hat{y}_i$ is the predicted tool-wear value.

The regression branch therefore models continuous tool degradation from the same integrated multimodal representation used for chip-morphology classification. This joint modelling structure allows the two related machining conditions to be evaluated within a common feature space while retaining task-specific output formulations.

2.10 Performance Evaluation Metrics

The proposed framework was evaluated using separate metrics for the classification and regression tasks. Classification performance was assessed primarily using accuracy, while tool-wear prediction was evaluated using RMS error (RMSE) and the coefficient of determination.

Classification accuracy was calculated as:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (10)$$

where, TP is true positive, TN is true negative, FP is false positive, and FN is false negative.

RMSE was calculated as:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (11)$$

The coefficient of determination was calculated as:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (12)$$

where, $\bar{y}$ represents the average tool wear. These metrics provide complementary measures of model performance. Classification accuracy quantifies the proportion of correctly identified chip-morphology samples, RMSE measures the magnitude of tool-wear prediction errors, and the coefficient of determination reflects the agreement between predicted and experimentally measured wear values.

To evaluate the contribution of the proposed integration structure, the hybrid multimodal framework was also compared with conventional machine-learning and deep-learning approaches, including RF, XGBoost, CNN-LSTM, and standalone Transformer models. In addition, the effects of different sensor combinations and architectural components were examined through multimodal fusion and ablation analyses. These comparisons were used to determine whether the coordinated integration of heterogeneous sensing, local feature extraction, bidirectional temporal modelling, and self-attention provided consistent performance benefits rather than improvements attributable to a single component.

3 Results and Discussion

The proposed hybrid multimodal modelling framework was evaluated for joint chip-morphology classification and tool-wear prediction under different CNC milling conditions. The experiments examined four aspects of model performance: the contribution of multimodal sensor integration, the accuracy of progressive flank-wear prediction, the discrimination of different chip morphologies, and robustness to variations in machining conditions. Architectural ablation experiments were further conducted to determine how the individual modelling components contributed to the performance of the integrated framework. All evaluations were based on synchronized vibration, AE, cutting-force, spindle-current, and temperature data acquired from the CNC milling process.

3.1 Multimodal Sensor Fusion Performance

Figure 2 compares the chip-morphology classification accuracy obtained with different sensor-modality configurations. Across the evaluated configurations, the proposed hybrid multimodal framework consistently achieved higher classification accuracy than the conventional Transformer and CNN–LSTM architectures. Single-sensor configurations, particularly those based on vibration, acoustic, or temperature signals alone, produced lower accuracy because each modality represented only part of the underlying machining dynamics. Force and vibration signals individually provided comparatively strong performance, which is consistent with their direct sensitivity to cutting mechanics, tool–workpiece interaction, and chip-formation behaviour.

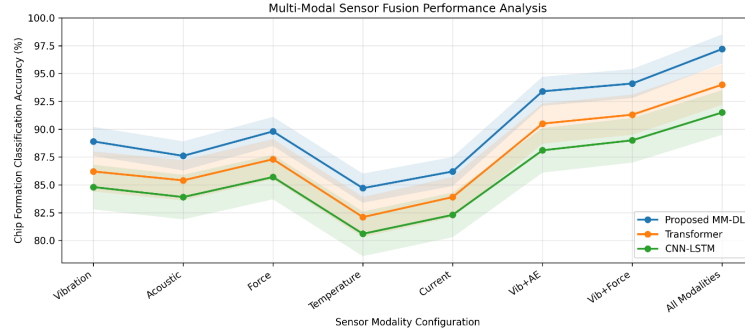


Figure 2. Classification performance of different sensor-modality configurations for chip-morphology analysis
Note: MM-DL: Multimodal deep learning; CNN: Convolutional neural network; LSTM: Long short-term memory; Vib: Vibration; AE: Acoustic emission.

Combining multiple sensing channels resulted in a clear improvement in classification performance. The vibration-plus-AE (Vib+AE) configuration achieved an accuracy of approximately 93.4%, while combining vibration and force (Vib+Force) increased the accuracy to approximately 94.1%. When all sensing modalities were integrated, the proposed framework achieved the highest classification accuracy, approximately 97.2

These results indicate that the performance gain was associated not simply with increasing the number of input signals, but with combining complementary information describing different physical aspects of the machining process. Mechanical vibration, acoustic activity, cutting load, spindle-current response, and thermal behaviour provide different but related observations of chip formation and tool–workpiece interaction. Their integration therefore produced a more complete representation of the machining state than any individual modality. The subsequent temporal and attention-based modelling stages were able to exploit this heterogeneous representation, supporting more stable discrimination among chip-morphology classes.

3.2 Tool-Wear Prediction Performance

Figure 3 compares the experimentally measured flank-wear values with those predicted by the proposed framework. The predicted values were closely distributed around the ideal regression relationship and showed strong agreement with the experimental observations. This distribution indicates that the integrated multimodal representation captured the progressive relationship between machining-sensor responses and tool degradation with relatively low prediction deviation.

The proposed framework also outperformed the baseline machine-learning approaches considered in the comparison. The close correspondence between measured and predicted wear suggests that heterogeneous sensing provides useful complementary information for continuous condition estimation. Unlike chip morphology, which is represented as a discrete state, flank wear evolves progressively during machining. Its estimation therefore depends on modelling changes in sensor responses over time rather than identifying an isolated signal pattern. The combination of local feature extraction, bidirectional temporal modelling, and long-range self-attention provides a structured mechanism for representing this evolution.

From an engineering perspective, reliable flank-wear estimation is relevant to dimensional accuracy, surface integrity, cutting-force stability, production efficiency, and tool-replacement planning. The observed agreement between experimental and predicted wear values therefore supports the use of the proposed hybrid framework for condition-based monitoring and predictive-maintenance applications in CNC milling.

3.3 Chip-Morphology Classification Analysis

The confusion matrix in Figure 4 provides a class-level assessment of the proposed framework for the four chip-morphology categories: continuous, segmented, discontinuous, and serrated chips. Most samples were concentrated along the diagonal of the matrix, indicating that the four chip states were generally distinguished correctly.

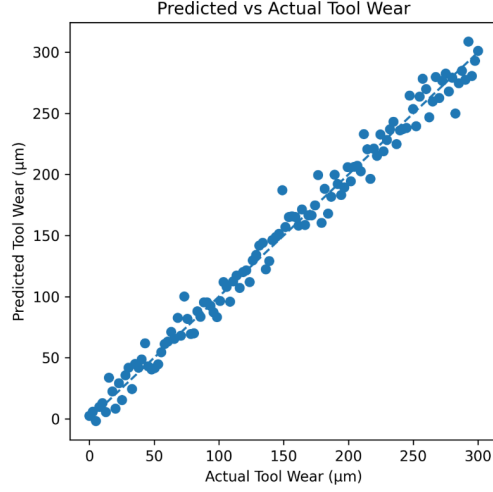


Figure 3. Comparison between experimentally measured and predicted tool-wear values using the proposed framework

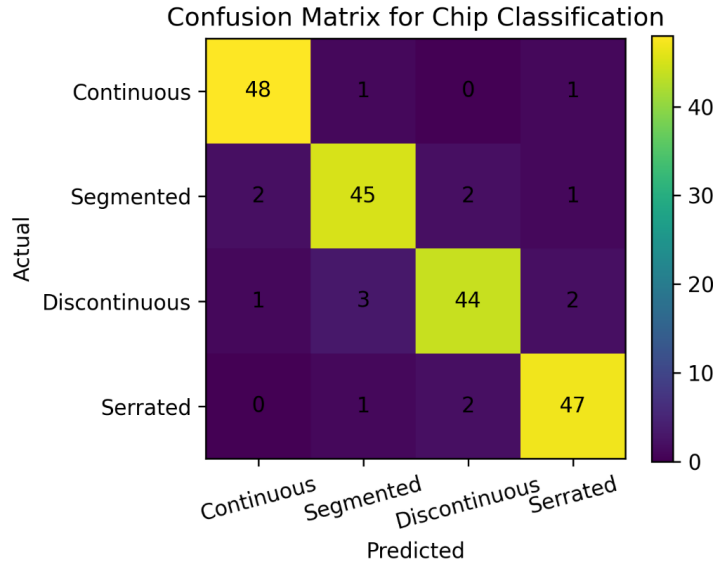


Figure 4. Confusion matrix for chip-morphology classification using the proposed hybrid multimodal framework

Continuous and serrated chips showed the strongest classification performance. Their comparatively distinctive signal characteristics in the frequency and force domains provide discriminative information that can be represented by the multimodal architecture. Most classification errors occurred between segmented and discontinuous chips. Under unstable cutting conditions, these two morphologies can generate partially overlapping transient responses, making their sensor representations less readily separable than those of continuous or serrated chips.

Importantly, the remaining confusion between these categories did not substantially reduce overall classification performance. The result suggests that the hybrid representation retained class-discriminative information even when different chip states exhibited partially similar local responses. CNN-based feature extraction captured localized characteristics of the machining signals, while the sequential and attention-based components incorporated their temporal context. The class-level results therefore provide further evidence that chip morphology benefits from modelling both instantaneous signal characteristics and their evolution within the machining sequence.

Accurate differentiation among chip morphologies is relevant to automated machining-quality assessment because changes in chip formation can reflect variations in cutting stability, material deformation, thermal conditions, and tool–workpiece interaction. The classification results consequently complement the continuous tool-wear estimates by providing a second representation of the evolving machining state.

3.4 Robustness Under Varying Cutting Conditions

The robustness of the proposed framework was evaluated under different machining conditions, including low-speed/high-feed, high-speed/low-feed, wet-milling, and dry-milling conditions. As shown in Figure 5, the proposed framework maintained comparatively stable prediction performance across the investigated operating conditions.

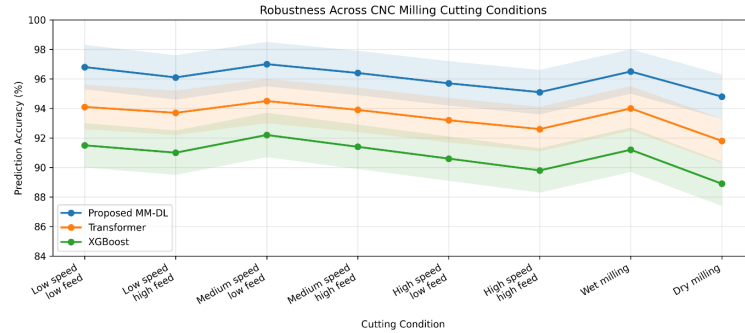


Figure 5. Robustness of the proposed framework under different computer numerical control (CNC) milling conditions

Note: MM-DL: Multimodal deep learning; XGBoost: eXtreme Gradient Boosting.

The conventional Transformer and XGBoost models showed greater performance degradation under more demanding conditions, particularly high-speed dry milling. In contrast, the prediction accuracy of the proposed framework remained above 94% throughout the evaluated conditions. This result is important because a monitoring architecture intended for practical machining systems must remain effective when operating parameters change rather than performing well only under a narrowly defined experimental configuration.

The observed robustness can be related to the integration of heterogeneous sensing information. When the informativeness or stability of an individual modality decreases, complementary information from the remaining sensing channels can preserve a useful representation of the machining state. The higher accuracy observed under wet-milling conditions was associated with lower thermal fluctuations and more stable chip evacuation. Dry milling, in contrast, produced greater signal variation as friction and thermal instability increased. Despite these differences, the integrated architecture maintained relatively stable performance under both lubrication conditions.

These findings indicate that the benefit of multimodal integration extends beyond improvements in nominal classification accuracy. The framework also showed reduced sensitivity to variations in operating conditions, which is a central requirement for transferring intelligent monitoring models from controlled experimental settings towards practical machining environments.

3.5 Ablation Study of the Hybrid Architecture

An ablation study was conducted to determine how the individual modelling components contributed to the performance of the complete architecture. The comparative results are presented in Figure 6.

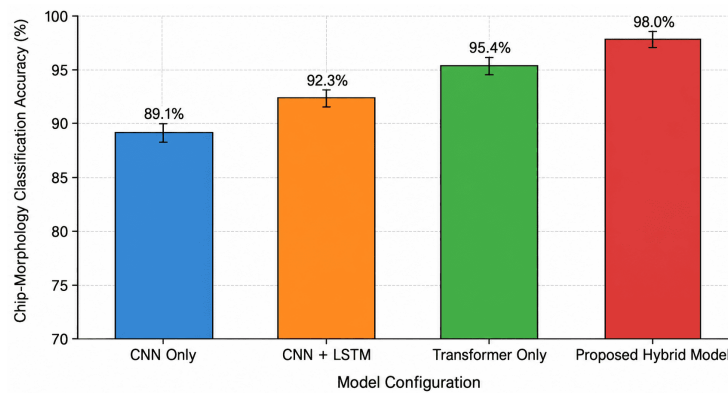


Figure 6. Ablation analysis of individual modelling components in the proposed hybrid architecture

Note: CNN: Convolutional neural network; LSTM: Long short-term memory.

The CNN-only configuration achieved approximately 89% prediction accuracy, indicating that localized convolutional features alone did not adequately represent the longer-range temporal dependencies present in the

machining signals. Incorporating recurrent temporal modelling increased the accuracy to approximately 92%, demonstrating the additional value of sequential information. The standalone Transformer achieved approximately 95%, further indicating that self-attention was effective in representing longer-range dependencies within the sensor sequences.

The complete hybrid multimodal framework achieved the highest accuracy, approximately 98%. The progressive improvement from the reduced configurations to the complete architecture indicates that the individual modelling components provided complementary rather than redundant functions. Convolutional processing represented localized signal characteristics, recurrent modelling incorporated sequential context, and Transformer-based attention captured interactions over longer temporal ranges. Multimodal fusion further provided these modelling components with information from multiple physical aspects of the machining process.

The ablation results therefore support the integration logic underlying the proposed architecture. Performance was not attributable solely to a single high-capacity learning component; rather, the highest accuracy was obtained when heterogeneous sensing, local representation learning, sequential modelling, and self-attention were combined. This finding is particularly relevant to the hybrid-modelling objective of the study because it provides empirical evidence that coordinated modelling components can offer advantages over isolated architectures.

3.6 Discussion

The experimental results collectively demonstrate that joint chip-morphology classification and progressive tool-wear prediction can benefit from the structured integration of heterogeneous sensing and complementary modelling mechanisms. The contribution of the proposed framework is therefore not limited to the use of multimodal data or to the introduction of an additional deep-learning component. Its principal characteristic is the coordination of sensor fusion, local feature extraction, sequential modelling, long-range attention, and task-specific prediction within a unified machining-monitoring architecture.

The sensor-fusion experiments provide the first indication of this effect. Classification accuracy increased from the individual and dual-sensor configurations to approximately 97.2% when all sensing modalities were used. This result suggests that vibration, AE, cutting force, spindle current, and temperature contain complementary information rather than interchangeable observations of the same process state. Their combination enables mechanical, acoustic, electrical, and thermal manifestations of machining behaviour to be represented simultaneously. Such heterogeneity is particularly relevant to intelligent engineering systems because the reliability of individual sensing channels can vary with operating conditions.

The architectural ablation analysis provides a second level of evidence. The increase from approximately 89% for the CNN-only configuration to approximately 92% with recurrent modelling, approximately 95% for the standalone Transformer, and approximately 98% for the complete framework indicates that performance improved as complementary representational mechanisms were incorporated. The result supports the methodological premise that local patterns, sequential evolution, and longer-range interactions represent different aspects of machining dynamics and should not necessarily be modelled by a single architecture. In this sense, the hybrid structure provides a coordinated representation across different temporal and feature scales.

The results obtained under varying cutting conditions further demonstrate the engineering relevance of this integration. Prediction accuracy remained above 94% across the investigated spindle-speed, feed-rate, and lubrication configurations, including dry-milling conditions associated with greater thermal and signal variability. This behaviour suggests that multimodal integration can reduce dependence on any single sensing channel and thereby improve the stability of the overall monitoring system when process conditions change. The observed robustness is particularly important for intelligent machining applications, where fixed operating assumptions are rarely maintained throughout production.

The two output tasks also provide complementary information about machining condition. Chip morphology represents discrete changes in material removal and cutting behaviour, whereas flank wear describes progressive tool degradation as a continuous process. Modelling both within the same multimodal representation establishes a common connection between sensor-level process observations and two related manifestations of machining state. The resulting architecture therefore extends beyond separate classification and regression models towards an integrated condition-monitoring structure.

Nevertheless, the present evaluation remains bounded by the machining conditions and sensing configuration considered in the experiments. Although the results demonstrate stable performance across the investigated spindle speeds, feed rates, and wet and dry conditions, broader validation across different machines, cutting tools, workpiece materials, sensor arrangements, and longer production cycles would be required to establish generalizability across industrial environments. Computational requirements for real-time implementation should also be examined explicitly when the framework is transferred from experimental evaluation to embedded or edge-based monitoring systems.

Overall, the findings indicate that the coordinated integration of heterogeneous sensing, convolutional feature

extraction, bidirectional temporal modelling, and Transformer-based self-attention provides an effective hybrid modelling strategy for intelligent CNC machining. The framework offers a common representation for chip-state recognition and tool-wear estimation while maintaining stable performance under varying operating conditions. This integration provides a methodological basis for predictive maintenance, adaptive machining control, tool-replacement planning, and real-time condition monitoring in intelligent manufacturing systems.

4 Conclusions

This study developed a hybrid multimodal modelling framework for the joint classification of chip morphology and prediction of progressive tool wear in CNC milling. The framework integrated vibration, AE, cutting force, spindle current, and temperature signals with complementary CNN-, BiLSTM-, and Transformer-based modelling mechanisms. The CNN captured localized signal characteristics, the BiLSTM represented bidirectional temporal dependencies, and Transformer-based self-attention modelled longer-range interactions within the fused sensor representation. This coordinated architecture enabled discrete chip-morphology classification and continuous flank-wear estimation to be performed within a common machining-state representation.

The experimental results demonstrated the value of both multimodal sensing and hybrid model integration. The full multimodal configuration achieved approximately 97.2% chip-morphology classification accuracy, outperforming configurations based on individual or reduced sensor combinations. The predicted flank-wear values showed close agreement with the experimental measurements, while the confusion-matrix analysis confirmed reliable discrimination among continuous, segmented, discontinuous, and serrated chip morphologies. Under variations in spindle speed, feed rate, and wet and dry milling conditions, the proposed framework maintained prediction accuracy above 94%. The ablation analysis further showed that the complete architecture achieved approximately 98% accuracy, compared with approximately 89% for the CNN-only configuration, 92% when recurrent temporal modelling was incorporated, and 95% for the standalone Transformer. These findings indicate that local feature extraction, sequential modelling, long-range attention, and heterogeneous sensor fusion provide complementary contributions to machining-state representation.

The results demonstrate that the principal benefit of the proposed framework lies in the structured integration of heterogeneous sensing and complementary modelling mechanisms rather than in any individual learning component. By representing mechanical, acoustic, electrical, and thermal responses within a unified architecture, the framework provides a common basis for monitoring two closely related manifestations of machining behaviour: chip formation and progressive tool degradation. This hybrid modelling strategy offers practical potential for condition-based maintenance, adaptive machining control, tool-replacement planning, and intelligent CNC process monitoring.

Further work should evaluate the framework across broader combinations of machine tools, cutting tools, workpiece materials, sensor configurations, and longer production cycles. Real-time computational performance and edge deployment also require systematic investigation before industrial implementation. Future extensions may additionally incorporate vision-based chip analysis and reinforcement-learning-based adaptive machining-parameter optimization to establish a more comprehensive closed-loop intelligent machining system.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The author declares no conflicts of interest.

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